# A candidate exact two-soliton S-matrix for the bad Boussinesq equation

Project *quantum_boussinesq_solitons* (Phase 2, next step 3)

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**Abstract**

We construct a candidate exact quantum two-soliton S-matrix for $v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx}$ and test it against everything Phase 2 established. The candidate is a ratio of Gamma functions in the *spectral-parameter difference* $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$, times a power of a cross-ratio of the cubic's roots, times the exact topological factor:

$$S_{\rm ov} = \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)}\Big(\frac{e^2 A_{12}}{\eta^2}\Big)^{i\eta} e^{\frac{i}{\hbar}\frac{2}{3}M_1M_2(v_1-v_2)}, \qquad S_{\rm ho} = \frac{\Gamma(1-i\eta)}{\Gamma(1+i\eta)}\Big(\frac{A_{12}\eta^2}{e^2}\Big)^{i\eta} e^{\frac{i}{\hbar}\frac{2}{3}M_1M_2(v_1-v_2)}.$$

It reproduces the semiclassical phase $\delta$ (including the topological term) up to a constant Maslov-type phase $\pm\pi/2$; it is exactly unitary, real-analytic, and obeys the crossing relation exactly (this fixes $S_{\rm ho}$). On the physical sheet its only poles at breather kinematics are simple poles exactly at the Bogdan–Charkina Bohr–Sommerfeld points $I_{\rm br} = N\hbar$, $N \geq 1$. Fused breather–soliton amplitudes are unitary with the right classical limit. Three other families are *excluded* at leading order: Gamma functions of root differences, any product $F(\beta_1-\beta_2)G(\beta_1+\beta_2)$ (so the pure $\mathfrak{sl}_3$ "mirror" form, and quantum dilogarithms of $\beta_1 \pm \beta_2$), and the pure exponential $e^{i\delta/\hbar}$ (no bound states). The candidate is a guess **(d)** whose properties are derived **(c)**; it is *not* derived from a quantum theory. Its remaining ambiguities are all $O(\hbar^0)$ in the phase and are listed with the input that would fix them.

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# 1 Setting and requirements

Conventions are those of `report.pdf`: $k = \sinh\beta$, $v = \epsilon\cosh\beta$, $\ell = \epsilon\cosh 3\beta$, $M = 2k$, $P = \frac{2}{3}\epsilon k^3\cosh\beta$; particle 1 starts on the left. The semiclassical phase is **(c)**

$$\delta(1,2) = \tfrac{2}{9}(\ell_1 - \ell_2)\log A_{12} + T_{12}, \qquad T_{12} = \tfrac{2}{3}M_1M_2(v_1 - v_2), \tag{1}$$

with $S \simeq e^{i\delta/\hbar}$. An exact candidate must (i) reproduce (1) as $\hbar \to 0$, (ii) be unitary, (iii) obey the crossing-type relation $S_{\rm ov}(\beta_1, \beta_2 + i\pi)S_{\rm ho}(\beta_1, \beta_2) = 1$ (the exact version of $\delta_{\rm ov}(\beta_1, \beta_2 + i\pi) = -\delta_{\rm ho}(\beta_1, \beta_2)$), (iv) have bound-state poles consistent with the string condition $\ell_1 - \ell_2 \in \mp\frac{9i}{2}\hbar\mathbb{N}$, equivalently with the Bogdan–Charkina (BC) spectrum $I_{\rm br} = N\hbar$ **(a)** [1], and (v) give consistent fused amplitudes. Scattering is diagonal, so Yang–Baxter is automatic.

**Roots of the cubic.** **(c)** (`scripts/smatrix_ansatz.py`) The three roots attached to a right-mover are

$$\mu_j = \tfrac{i}{\sqrt{3}}\cosh\!\big(\beta + \tfrac{i\pi(2j+1)}{3}\big), \quad j = -1, 0, 1,$$

roots of $\mu^3 + \frac{1}{4}\mu = \frac{i}{12\sqrt{3}}\ell$; the first two are the pair $\frac{1}{2}(iv/\sqrt{3} \pm k)$ of Phase 2, whose cross-ratio is $A_{12}$. Consequently, for every root $\mu_{1a}$ of soliton 1,

$$\prod_b (\mu_{1a} - \mu_{2b}) = \tfrac{i}{12\sqrt{3}}(\ell_1 - \ell_2). \tag{2}$$

So $\ell_1 - \ell_2$ is itself a product of root differences, and every function below is a product of powers of root differences.

# 2 Families tested and excluded

Single-particle factors are a convention, so we compare the mixed derivative $\partial_{\beta_1}\partial_{\beta_2}$ of the leading phase with that of $\delta$, by least squares at 40 random overtaking points (`scripts/smatrix_rejected_forms.py`). **(c)**

| Family | Leading phase spans | Residual |
|---|---|---|
| $\prod\Gamma(1+ic\,x_{ab}/\hbar)^{n_{ab}}$, $x_{ab} = \mu_{1a} - \mu_{2b}$ (any complex $c, n$), $\times e^{iT/\hbar}$ | $\Re, \Im\; x\log x$, $x$, $T$ | $2 \times 10^{-2}$: excluded |
| $F(\beta_1 - \beta_2)\,G(\beta_1 + \beta_2)$: $\mathfrak{sl}_3$ Toda / sine-Gordon mirror blocks, quantum dilogarithms of $\beta_1 \pm \beta_2$ | $f(D) + g(S)$, so $\partial_S\partial_D = 0$ | $\partial_S\partial_D\delta \neq 0$, e.g. $-0.55$ at $(S, D) = (1.4, 0.6)$: excluded |
| $e^{i\delta/\hbar}$ exactly | $\delta$ | passes (i)–(iii), *no poles*: excluded by (iv) |
| Control: $\Gamma$ in $\eta$ (Sec. 3) | $(\ell_1 - \ell_2)\log x_{ab}$, $T$ | $5 \times 10^{-16}$ |

The obstruction to the first two is the same: $\delta$ has the *uniform* weight $\ell_1 - \ell_2$ in front of every logarithm, while Stirling asymptotics of $\Gamma(\ldots + iX/\hbar)$ give weight $X$ in front of $\log X$. The weight must therefore be the Gamma argument. This forces the Gamma functions to depend on $\eta \propto \ell_1 - \ell_2$ only. **(c)**The quantum-dilogarithm case is excluded only for arguments that are functions of $\beta_1 \pm \beta_2$ separately; a dilogarithm with mixed arguments was not tested.

## 3 The candidate

Let $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ and $\mathcal{A}_{12} = \dfrac{81 A_{12}}{4(\ell_1 - \ell_2)^2}$, so $A_{12}/\eta^2 = \hbar^2 \mathcal{A}_{12}$. **(d)**

$$S_{\epsilon\epsilon}(1,2) = \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)} \left(e^2 \hbar^2 \mathcal{A}_{12}\right)^{i\eta} e^{iT_{12}/\hbar} \qquad (\epsilon_1 = \epsilon_2), \tag{3}$$

$$S_{+-}(1,2) = \frac{\Gamma(1-i\eta)}{\Gamma(1+i\eta)} \left(\frac{A_{12}\eta^2}{e^2}\right)^{i\eta} e^{iT_{12}/\hbar} \qquad \text{(head-on)}. \tag{4}$$

**Proposition 3.1** (**(c)**, `scripts/smatrix_ansatz.py`). *1. $\mathcal{A}_{12}$ is finite and nonzero at threshold $\beta_1 = \beta_2$, so (3) has no singularity there: the $\log(\ell_1 - \ell_2)^2$ singularity of $\log A_{12}$ is carried by the Gamma ratio. Using (2), $\mathcal{A}_{12}$ is a ratio of products of root differences.*

*2. $\log \mathcal{A}_{12} = h(\beta_1 + \beta_2) + g(\beta_1 - \beta_2)$: the $\mathfrak{sl}_3$/sine-Gordon blocks of Phase 2 (Prop. 5.1 of `report.pdf`) survive, but only* in the exponent*, multiplied by $\eta$.*

*3. $\Gamma(ix)/\Gamma(-ix) = -\Gamma(1+ix)/\Gamma(1-ix)$. Hence the Gamma part is that of the quantum Toda two-body amplitude $R = \Gamma(i\kappa)/\Gamma(-i\kappa)\hbar^{2i\kappa}$ (`toda_route.pdf`), with $\kappa = \Delta p/\hbar \mapsto \eta$, and the Toda Lax eigenvalue $p \mapsto \frac{2}{9}\ell$. That $p$ maps to $\ell$ in the phase weight was found in Phase 2b; that the Gamma structure survives the Boussinesq limit is* **(d)**.

## 4 Tests

### 4.1 Semiclassical limit, unitarity, crossing

**Proposition 4.1** (**(c)**, `scripts/smatrix_semiclassical_unitarity_crossing.py`). *For real rapidities, at three overtaking and three head-on points:*

1. $\arg S - \delta/\hbar \to \frac{\pi}{2}\sigma \operatorname{sgn}\eta$ *(mod $2\pi$), $\sigma = +1$ (same direction), $-1$ (head-on), with remainder $O(\hbar)$; at $\hbar = 10^{-2}, 10^{-3}, 10^{-4}$ the remainder is e.g. $-6.4\times10^{-4}, -6.4\times10^{-5}, -6.4\times10^{-6}$. Without the factor $e^{iT/\hbar}$ the remainder is $O(1/\hbar)$, so the topological term is required.*
2. *$|S| = 1$ and $S(1,2)S(2,1) = 1$ to 30 digits.*
3. *$S_{\mathrm{ov}}(\beta_1, \beta_2 + i\pi)\, S_{\mathrm{ho}}(\beta_1, \beta_2) = 1$ to $10^{-27}$; at leading order this is $\delta_{\mathrm{ov}}(\beta_1, \beta_2 + i\pi) + \delta_{\mathrm{ho}} = 0$.*
4. *$S(\bar\beta_1, \bar\beta_2) = 1/\overline{S(\beta_1, \beta_2)}$ at generic complex rapidities.*

Crossing is what fixes (4): under $\beta_2 \to \beta_2 + i\pi$, $\eta_{\mathrm{ov}} \to \eta_{\mathrm{ho}}$ and $A_{\mathrm{ov}} \to 1/A_{\mathrm{ho}}$ (Phase 2), so the inverse Gamma ratio and the power $(A\eta^2/e^2)^{i\eta}$ are forced. Using (3) for head-on scattering too would violate crossing. During the work we first wrote (4) with $(\eta^2/e^2 A)^{i\eta}$; the semiclassical test failed, and crossing gave the correct form.

The constant $\pm\pi/2$ is a Maslov-type phase at $O(\hbar^0)$. The report listed a missing Maslov phase among its least-sure points; the candidate predicts one. **(d)**

### 4.2 Poles and the breather spectrum

Take two right-movers at $\beta_{1,2} = a \mp ib$ with $\pi/3 < b < \pi/2$ (right-moving breathers; left-moving ones are their parity images).

**Proposition 4.2** (**(c)**, `scripts/smatrix_poles_bootstrap.py`). *1. With this labelling $\mathrm{Im}(P_1 - P_2) > 0$: the relative momentum lies in the upper half plane, so this is the physical sheet for bound states of $S(1,2)$. Total $E$ and $P$ are real.*

2. *$i\eta = \frac{4}{9}\sinh 3a \sin 3b/\hbar = -I_{\rm br}/\hbar$, with $I_{\rm br} = -\frac{4}{9}\sinh 3a \sin 3b$ the BC adiabatic invariant. Hence $S(1,2)$ has simple poles (from $\Gamma(1+i\eta)$) exactly at $I_{\rm br} = N\hbar$, $N = 1, 2, \ldots$: the BC Bohr–Sommerfeld spectrum, with their $N$ (not $N + \frac{1}{2}$). Verified for $N \leq 4$ at $a = 0.5, 0.8$, $\hbar = 0.1$ ($|S||\delta b|$ constant to $10^{-9}$).*

3. *There is no pole at $N = 0$ (threshold $b = \pi/3$); there the Toda form $\Gamma(i\eta)$ would have one. This selects $\Gamma(1 + i\eta)$.*

4. *The form that is regular at real threshold and the form with poles on the physical sheet coincide. This was not imposed and is a non-trivial consistency check.*

5. *At a pole $e^{i\eta \cdot 2\pi i} = 1$, so the residue modulus does not depend on the branch of $\log \mathcal{A}_{12}$ (which lies on its cut, $\mathcal{A}_{12} < 0$, at breather kinematics). The residue phase does.*

Sample energies $E_N = E(\beta_1) + E(\beta_2)$ at $\hbar = 0.1$, $a = 0.8$: $0.8711, 0.9759, 1.0759, 1.1705$ for $N = 1..4$ (**(c)**, numerical; identical to BC's quantisation because the conditions coincide).

### 4.3 Bootstrap: breather–soliton amplitudes

**Proposition 4.3** (**(c)**, `scripts/smatrix_poles_bootstrap.py`)**.** *Define $S_{B_N,3} = S(1,3)S(2,3)$ at the $N$-th pole. Then:*

1. *$|S_{B_N,3}| = 1$ for real $\beta_3$ of either direction, and $\delta_{13} + \delta_{23}$ is real (the classical breather–soliton phase, by Hirota additivity).*

2. *$\arg S_{B_N,3} - (\delta_{13} + \delta_{23})/\hbar \to \pi$ as $\hbar = I_{\rm br}/N \to 0$ at fixed $(a, b)$: two Maslov-type constants.*

3. *With $x_c = \frac{i}{2}(\eta_{13} + \eta_{23})$, the Gamma part telescopes to*

$$\frac{\Gamma(1 + x_c - \frac{N}{2})\Gamma(1 + x_c + \frac{N}{2})}{\Gamma(1 - x_c + \frac{N}{2})\Gamma(1 - x_c - \frac{N}{2})}.$$

*It has a simple pole at $x_c = -\frac{N}{2}$, double poles at $x_c = -\frac{N}{2} - 1 - m$, and the mirror-image zeros; $N - 1$ pole–zero pairs cancel in between. This was checked numerically for $N = 1, 2, 3$.*

*Interpretation* **(d)**: the simple pole is at $\ell_3 = \ell_2$, $\ell_1 - \ell_3 = N$-string, an exchange of soliton 3 with a constituent. The double poles extend the string, $\ell_3 = \ell_2 - \frac{9i}{2}(m + 1)\hbar$, and would need a Coleman–Thun-type explanation. Whether these lie on the physical sheet of the three-body kinematics was *not* determined, so the bootstrap is only partially tested. Breather–breather amplitudes follow by the same fusion but were not analysed.

## 5 Summary of tests

| Requirement | Result for (3)–(4) | Status |
|---|---|---|
| Semiclassical limit | $\delta/\hbar \pm \pi/2 + O(\hbar)$, topological term needed | **(c)** passed |
| Unitarity | exact | **(c)** passed |
| Crossing | exact; fixes the head-on form | **(c)** passed |
| Real analyticity | exact | **(c)** passed |
| Poles vs string / BC | simple poles exactly at $I_{\rm br} = N\hbar$, physical sheet, none at $N = 0$ | **(c)** passed |
| Bootstrap | fused amplitudes unitary, right classical limit; pole interpretation open | **(c)** partial |
| Derivation from a quantum theory | none | **(d)** |

## 6 Remaining ambiguities and what would fix them

All tests constrain the phase only at orders $\hbar^{-1}$ (fixed by $\delta$) and at the pole positions. The following modifications pass every test above, or fail only the poles in a way the present data cannot decide. **(c)** (that they pass) / **(d)** (that this list is complete)

1. **CDD factors.** Any $f(1,2)$ with $f(1,2)f(2,1)=1$, $|f|=1$ for real rapidities, $f_{\rm ov}(\beta_1,\beta_2+i\pi)f_{\rm ho}(\beta_1,\beta_2)=1$, and $\log f = O(\hbar^0)$. Examples: $\prod_j \frac{\sinh\frac{\pi}{2}(\eta-i\alpha_j)}{\sinh\frac{\pi}{2}(\eta+i\alpha_j)}$, or $\frac{c+i\eta}{c-i\eta}$ with constant $c$. They change one-loop phases and add or remove poles; the latter could be tested against a quantum breather spectrum.

2. **Renormalisation of** $\hbar$ in $\eta$, $\hbar \to \hbar_{\rm ren} = \hbar(1+c_1\hbar+\ldots)$, as in sine-Gordon $\beta^2 \to \beta^2/(1-\beta^2/8\pi)$. This shifts the breather spectrum at $O(\hbar^2)$.

3. **Shift of the string**: $\Gamma(\frac{1}{2}+i\eta)$ instead of $\Gamma(1+i\eta)$, i.e. $I_{\rm br}=(N-\frac{1}{2})\hbar$. This is excluded only if one trusts BC's integer quantisation, which is itself a leading-order Bohr–Sommerfeld statement.

4. $O(\hbar)$ **corrections to** $\mathcal{A}_{12}$ inside the power, and the branch of $\log\mathcal{A}_{12}$ at complex rapidities (residue phases).

5. **The Maslov-type constant** $\pm\frac{\pi}{2}$, and whether the overall sign of $S$ is a convention.

**Input that would fix them.**

- *One-loop soliton and breather spectrum* (DHN-type; next step 2 of the report). The $O(\hbar^0)$ correction to the breather energies $E_N$ fixes items 2–3. The one-loop two-soliton phase (from the phonon phase shifts on the reflectionless soliton background) fixes item 4 and tests the $\pm\pi/2$.

- *Small-amplitude Toda limit* (next step 1: the Fermi-edge double scaling $\beta\to 0$, $\hbar\propto\beta^2$, of the quantum Toda Bethe ansatz). This gives an exact quantum S-matrix in the corner $\beta\to 0$ against which (3)–(4), including possible CDD factors, can be compared directly. Note $\eta=O(1)$ there means $\hbar\propto\beta^4$ for overtaking ($\ell_1-\ell_2\approx\frac{9}{2}(\beta_1^2-\beta_2^2)$) but $\hbar=O(1)$ for head-on ($\ell_1-\ell_2\approx 2$). So the head-on Gamma ratio is never semiclassical in that corner. **(c)** (elementary)

- *A quantum lattice regularisation* at the dispersion scale would be needed to derive the candidate rather than guess it; none is known (`toda_route.pdf`).

## A Scripts

All are in `phase2/scripts`; each ends with `ALL CHECKS PASSED`.

- `smatrix_ansatz.py`: definitions; roots, Eq. (2), Prop. 3.1.
- `smatrix_semiclassical_unitarity_crossing.py`: Prop. 4.1.
- `smatrix_poles_bootstrap.py`: Props 4.2–4.3.
- `smatrix_rejected_forms.py`: Section 2.

*Failures along the way:*

- The first root labelling mismatched the Phase 2 pair. It was fixed by matching to `consistency_checks.py`.

- The first head-on form had the wrong sign of the $A$-power; it failed the semiclassical and crossing tests and was corrected as described above.
- A threshold-size tolerance was set too tightly ($|\mathcal{A}| > 10^{-3}$, whereas $\mathcal{A} \approx 2 \times 10^{-4}$ at $\beta = 1.6$).
- $|S|$ at complex breather kinematics is very large ($\sim 10^{23}$ at $a = 0.8$) because $|A^{i\eta}| \neq 1$ there. The pole test therefore uses scaling with the distance to the pole, not absolute size.

# References

[1] M. M. Bogdan, O. V. Charkina, arXiv:2502.14695, *Low Temp. Phys.* 51 (2025).